

# TASK-3

The relay should be turned on when the ultrasonic sensor measures a distance within 30 cm ( Threshold )

## The Wiring Guide

| Ultrasonic Sensor (HC-SR04) | ESP32 WROOM Pin | Notes                                      |
|-----------------------------|-----------------|--------------------------------------------|
| <b>VCC</b>                  | 5V / VIN        | Power for the sensor                       |
| <b>GND</b>                  | GND             | Ground                                     |
| <b>Trig</b>                 | GPIO 5          | Output from ESP32 to trigger pulse         |
| <b>Echo</b>                 | GPIO 18         | Input to ESP32 (Use voltage divider here!) |

| Relay Module | ESP32 WROOM Pin  | Notes                              |
|--------------|------------------|------------------------------------|
| <b>VCC</b>   | 5V / VIN or 3.3V | Depending on your relay board spec |
| <b>GND</b>   | GND              | Ground                             |
| <b>IN</b>    | GPIO 19          | Controls the relay switch          |

## MicroPython Code

```
import machine
import time

# --- Configuration ---
TRIGGER_PIN = 5
ECHO_PIN = 18
RELAY_PIN = 19
THRESHOLD_CM = 30.0 # Distance threshold in cm

# --- Pin Initialization ---
trig = machine.Pin(TRIGGER_PIN, machine.Pin.OUT)
echo = machine.Pin(ECHO_PIN, machine.Pin.IN)
relay = machine.Pin(RELAY_PIN, machine.Pin.OUT)

# Ensure relay starts OFF
# Note: Active-Low relays turn ON when output is 0. Active-High turn ON when 1.
# If your relay behavior is inverted, change this to 0.
relay.value(0)

def get_distance():
```

```

"""Measures distance using the HC-SR04 ultrasonic sensor."""
# Ensure trigger is low
trig.value(0)
time.sleep_us(2)

# Send a 10 microsecond pulse to trigger
trig.value(1)
time.sleep_us(10)
trig.value(0)

# Measure the duration of the echo pulse
duration = machine.time_pulse_us(echo, 1, 30000) # 30ms timeout

if duration < 0:
    # If timeout occurs (return value -1 or -2)
    return None

# Calculate distance in cm
# Speed of sound is ~343 m/s or 0.0343 cm/us.
# Pulse goes there and back, so we divide by 2.
distance = (duration * 0.0343) / 2
return distance

print("Ultrasonic Relay Controller Initialized...")

# --- Main Loop ---
while True:
    dist = get_distance()

    if dist is not None:
        print(f"Distance: {dist:.2f} cm")

        # Check if an object is within the threshold
        if dist <= THRESHOLD_CM:
            print("Target detected! Turning Relay ON.")
            relay.value(1) # Set to 0 if using an Active-Low relay module
        else:
            print("Clear. Turning Relay OFF.")
            relay.value(0) # Set to 1 if using an Active-Low relay module
        else:
            print("Out of range or sensor error.")
    # Wait 500ms before the next reading to avoid flooding the sensor
    time.sleep(0.5)

```

### NOTE:

**THRESHOLD\_CM =30 (Add this in the code)**

# OUTPUT

```
Thonny - C:\Users\ANAGHA\main.py @ 49:5
File Edit View Run Tools Help

main.py
44 print("Ultrasonic Relay Controller Initialized...")
45
46 # --- Main Loop ---
47 while True:
48     dist = get_distance()
49     |
50     if dist is not None:
51         print(f"Distance: {dist:.2f} cm")
52
53     # Check if an object is within the threshold
54     if dist < THRESHOLD_CM:

Shell
Clear. Turning Relay OFF.
Distance: 391.88 cm
Clear. Turning Relay OFF.
Distance: 72.80 cm
Clear. Turning Relay OFF.
Distance: 392.75 cm
Clear. Turning Relay OFF.
Distance: 392.32 cm
Clear. Turning Relay OFF.
Distance: 392.73 cm
Clear. Turning Relay OFF.
Distance: 392.39 cm
Clear. Turning Relay OFF.
Distance: 391.93 cm
Clear. Turning Relay OFF.
Distance: 391.91 cm
Clear. Turning Relay OFF.
Distance: 392.31 cm
Clear. Turning Relay OFF.
Out of range or sensor error.
***** of range or sensor error.
Distance: 7.53 cm
Target detected! Turning Relay ON.
Distance: 12.26 cm
Target detected! Turning Relay ON.
Distance: 43.
```

MicroPython (ESP32) • CP2102 USB to UART Bridge Controller @ COM3

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